

Class 10: Structural Bioinformatics 1

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PBD Statistics

The Protein Data Bank (PDB) is the main repository of biomolecular structures. Let's see what it contains:

```
stats <- read.csv("Data Export Summary.csv")
stats
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	178,795	21,825	12,773	343	226
2	Protein/Oligosaccharide	10,363	3,564	34	8	11
3	Protein/NA	9,106	6,335	287	24	7
4	Nucleic acid (only)	3,132	221	1,566	3	15
5	Other	175	25	33	4	0
6	Oligosaccharide (only)	11	0	6	0	1

	Neutron	Other	Total
1	84	32	214,078
2	1	0	13,981
3	0	0	15,759
4	3	1	4,941
5	0	0	237
6	0	4	22

```
stats$X.ray
```

```
[1] "178,795" "10,363" "9,106" "3,132" "175" "11"
```

```
sum(stats$Neutron)
```

```
[1] 88
```

The comma in these numbers leads to the numbers here being read as character.

```
c(100, 10, "jolie")
```

```
[1] "100"  "10"   "jolie"
```

```
library(readr)
stats <- read_csv("Data Export Summary.csv")
```

Rows: 6 Columns: 9

-- Column specification -----

Delimiter: ","

chr (1): Molecular Type

dbl (4): Integrative, Multiple methods, Neutron, Other

num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
stats
```

A tibble: 6 x 9

	`Molecular Type`	`X-ray`	EM	NMR	Integrative	`Multiple methods`	Neutron
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Protein (only)	178795	21825	12773	343	226	84
2	Protein/Oligosacch~	10363	3564	34	8	11	1
3	Protein/NA	9106	6335	287	24	7	0
4	Nucleic acid (only)	3132	221	1566	3	15	3
5	Other	175	25	33	4	0	0
6	Oligosaccharide (o~	11	0	6	0	1	0

i 2 more variables: Other <dbl>, Total <dbl>

```
n.xray <- sum(stats$`X-ray`)
```

```
#n.em <-
```

```
n.total <- sum(stats$Total)
```

```
n.xray/n.total
```

```
[1] 0.8095077
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

X-ray = 81%

EM = 13%

```
n.xray/n.total
```

```
[1] 0.8095077
```

```
n.em <- sum(stats$EM)
n.em/n.total
```

```
[1] 0.1283843
```

Q2: What proportion of structures in the PDB are protein?

```
rowsum <- sum(stats[2, c(2:9)])
rowsum/sum(stats$Total)
```

```
[1] 0.1122891
```

11.2% is the proportion of structures in the PDB are protein.

Q3: SKIP

Visualizing the HIV-1 protease structure

We can use the Molstar viewer online: <https://molstar.org/viewer/>

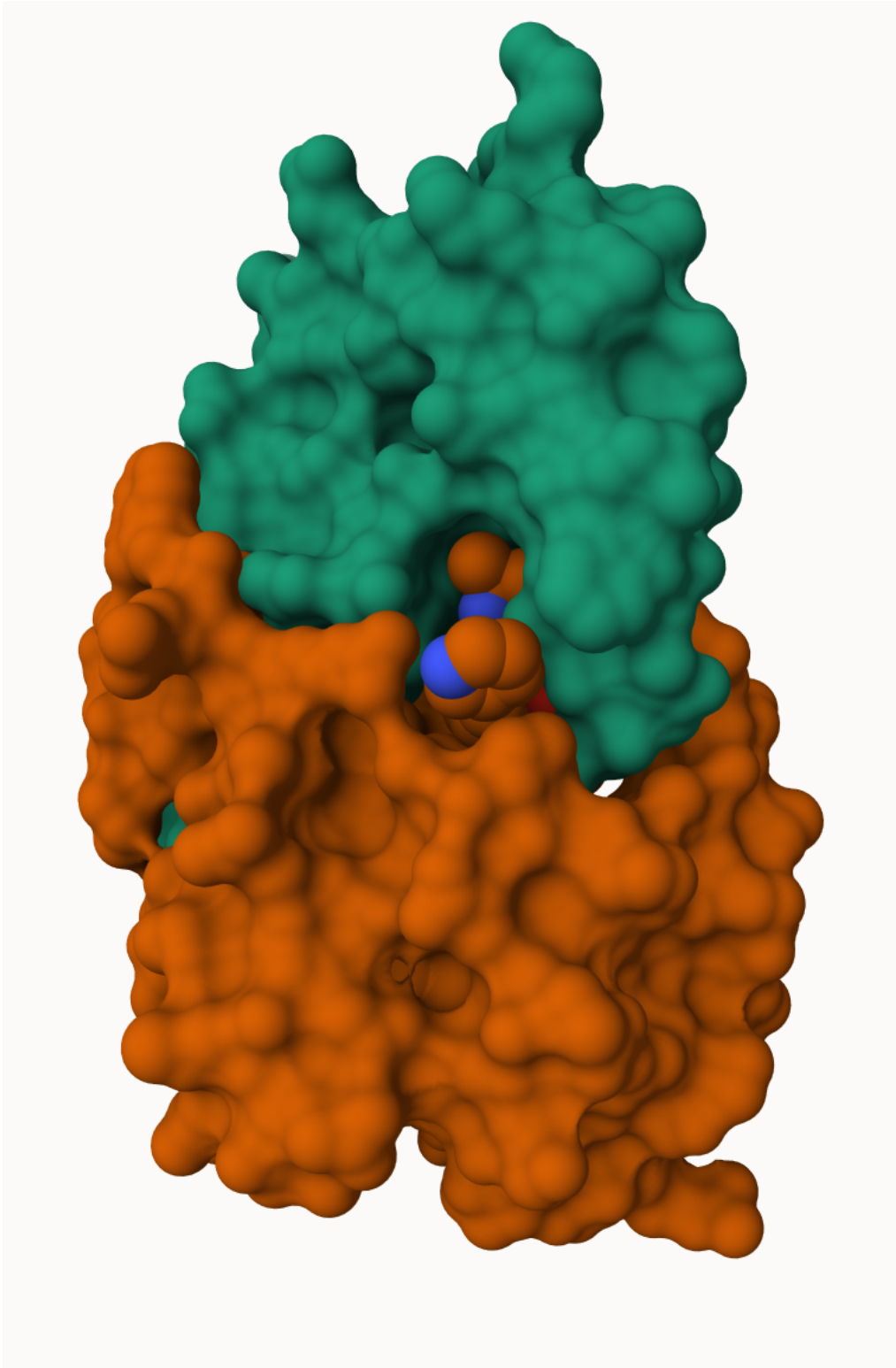


Figure 1: My first image of HIV-Pr with surface display showing ligand binding

A new clean image showing the catalytic ASP25 amino acids in both chains of the HIV-PR dimer along with the inhibitor and all important active site water.



Bio3D packages for structural bioinformatics

```
library(bio3d)
pdb <-read.pdb("1HSG.png")
pdb
```

Call: read.pdb(file = "1HSG.png")

Total Models#: 1

Total Atoms#: 0, XYZs#: 0 Chains#: 0 (values:)

Protein Atoms#: 0 (residues/Calpha atoms#: 0)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 0 (residues: 0)

```
Non-protein/nucleic resid values: [ none ]
```

```
+ attr: atom, xyz, calpha, call
```

```
head(pdb$atom)
```

```
[1] type   eleno  elety  alt    resid  chain  resno  insert x      y  
[11] z      o      b      segid  elesy   charge  
<0 rows> (or 0-length row.names)
```

```
# library(bio3dview)  
# view.pdb(pdb)
```

```
# Select the important ASP 25 residue  
sele <- atom.select(pdb, resno=25)  
  
# and highlight them in spacefill representation  
#view.pdb(pdb, cols=c("navy","orange"),  
#         highlight = sele,  
#         highlight.style = "spacefill")
```

Predicting functional motions of a single structure

Read an ADK structure from the PBD database:

```
# adk <- read.pdb("6s36")  
# adk
```

```
#m <- nma(adk)  
#plot(m)
```

write out our results as a wee trajectory/movie of a predicted

```
# mktrj(m, file="adk_m7.pdb")
```